MIL–T–945A
AMENDMENT—2
14 MAY 1953

SUPERSEDING
AMENDMENT—1
18 December 1951

MILITARY SPECIFICATION

TEST-EQUIPMENT, FOR USE WITH ELECTRONIC EQUIPMENT: GENERAL SPECIFICATION

This amendment forms a part of Military Specification MIL–T–945A, dated 7 March 1950, and has been approved by the Departments of the Army, the Navy, and the Air Force

Page 1, paragraph 2.1: Delete reference to Military Specifications JAN–1 and JAN–I–6, and add:

"MIL–E–1—Electron Tubes.
MIL–M–6—Meters, Electrical-Indicating, Panel Type.
MIL–M–10304—Meters, Ruggedized, Electrical.
MIL–E–15090—Enamel, Equipment, Light-Gray (Formula No. 111)."

Pages 2 and 3, paragraph 2.1: Delete reference to Navy Department Specifications 17E11 and 52E14.

Page 3, paragraph 2.2, lines 5 to 15, inclusive: Delete and substitute:

"STANDARDS
MILITARY
JAN–STD–15—Electrical and Electronic Symbols.
MIL–STD–108—Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL–STD–125—Standard Guides for Preparation of Item Descriptions.
MIL–STD–200—Electron Tubes.
PUBLICATIONS
MILITARY
Armed Services Index of R. F. Transmission Lines and Fittings."

Section 3: Change "Index of Army-Navy R–F Transmission Lines and Fittings" to "Armed Services Index of R. F. Transmission Lines and Fittings" wherever it appears.

Page 8: Add as new paragraph:

"3.5.3 Waveguide flanges, assemblies and adapters shall be selected in accordance with the Armed Services Index of R. F. Transmission Lines and Fittings and shall be in accordance with the specifications and drawings referenced therein."

Page 9, paragraph 3.9.2, second and last lines: Change "JAN–1" to "MIL–E–1" and "the Preferred List of Army-Navy Electron Tubes" to "Standard MIL–STD–200".

Page 12, paragraph 3.18.1: Delete and substitute:

"3.18.1 Meters employed in test equipment shall be sealed meters in accordance with Specification MIL–M–6. Where ruggedized meters are specified in the individual equipment specification, meters shall conform to Specification MIL–M–10304. Meters shall be properly bonded to the panel and shall use a conducting glass window or coating material where necessary to comply with the noise suppression requirements."

Pages 15 and 16, paragraphs 3.30.1 through 3.30.6: Delete and substitute:

"3.30.1 Electron tubes shall be in accordance with Specification MIL–E–1.

"3.30.1.1 The number of tubes and types of tubes used in each particular test set shall be

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kept to a minimum. Electron tubes used in the design of each particular test set shall be chosen from Standard MIL–STD–200 except that the following types shall be used in lieu of the prototypes listed in Standard MIL–STD–200: OA2WA, OB2WA, 6AU6WA, 12AT7WA and 5814A. Use of any tubes not shown in Standard MIL–STD–200 will be acceptable only if satisfactory to the bureau or agency concerned. Before consideration will be given to acceptance of a nonpreferred type, the following information is required:

(a) Quantitative data on the electrical advantages of the proposed tube over the nearest equivalent type on the preferred list.

(b) Proof that the proposed tube is mechanically adequate for the service intended.

(c) Submission of tube test data and performance limits of the tube which would have to be maintained by the tube manufacturer to insure satisfactory performance of the test set.

(d) Information on production availability of the tube.

"**3.30.1.2** The contractor is not authorized to develop any new tubes necessary for the test set, but is required to notify the bureau or agency concerned should the need for such development arise.

"**3.30.1.3** *Design requirements.*—Test equipment shall be so designed that the specified performance will be obtained from tubes listed in Standard MIL–STD–200, meeting and operating within the requirements of Specification MIL–E–1. Test sets and the electron tube circuits therein shall be so designed that, when a tube with any practical combination of tube characteristics controlled by Specification MIL–E–1 is used (all other tubes being average), the test set will meet the performance standards specified in the individual test set specification. In general, the operation of a circuit shall not depend on an electron tube characteristic not controlled by Specification MIL–E–1.

"**3.30.1.4** *Design information.*—The contractor shall supply any detailed design information concerning the specific application of electron tubes that may be requested by the bureau or agency concerned.

"**3.30.1.5** *Tube complement reports.*—All tube complements, even though tentative, shall be reported by the contractor to the bureau or agency concerned as soon as determined; all changes in such tube complements shall also be reported as they occur, during the course of the contract. The bureau or agency concerned shall be advised promptly of the final tube complement as soon as it has been determined."

Page 18, paragraph 3.35.1: Delete and substitute:

"**3.35.1** *Finishes of transit, combination, and instrument case.*—External surfaces shall be finished with gray enamel conforming to Specification MIL–E–15090. Class 2 enamel shall be used if weather protected, and class 1 enamel shall be used if weather exposed. Where it is not practical to use enamel, the color of the finish shall conform to Specification MIL–E–15090."

Page 20, paragraph 3.37, last two lines: Delete "Publication JANP–109" and substitute "Standard MIL–STD–125."

Page 21, paragraph 3.38.4.5: Delete and substitute:

"**3.38.4.5** When the item or part is of metal, the marking shall be etched, cast, or engraved. A label as specified in 3.38.4.10 or a stamp with waterproof ink may be used only on internal parts where the above methods are impractical."

Page 22, paragraph 3.39.1, first sentence: Delete and substitute "The diagrams shall contain the actual values of the parts used, such as resistors in ohms and capacitors in microfarads."

Page 25, paragraphs 3.43.1.3 (b) and (c): Delete and substitute:

"(b) Radio Test Set AN/PRM–1; over the frequency range of 0.150 to 15 megacycles.

"(c) Noise-Field Intensity Meter TS–587/U; over the frequency range of 15 to 400 megacycles."

Page 25, paragraph 3.45: Add "The equipment shall be turned ON by a clockwise, upward, or to the right movement of the switch."

Page 27, paragraphs 3.50.1 and 3.50.2: Delete and substitute:

"**3.50.1** *Transit cases and combination cases.*—Transit cases and combination cases

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shall be watertight as specified in Standard MIL–STD–108.

"3.50.2 *Instrument cases.*—Instrument cases shall be dripproof protected in accordance with Standard MIL–STD–108."

Page 31, paragraph 4.2.3 (c), last line: Change "JAN–1" to "MIL–E–1."

Page 33, paragraph 4.4.5: Delete and substitute:

"4.4.5 *Vibration test.*—The test set, with and without its transit case or in its combination case, shall be mounted on the vibration table. If internal batteries are required, they shall be placed in the case and fastened in a normal manner required for operation. During the vibration test, the test set shall be energized and monitored except when installed in its transit case."

Page 34, paragraph 4.4.6.1, third sentence: Delete and substitute: "Portable test sets shall be secured to the test machine by means of canvas straps."

Page 34: Add the following:

"4.4.7 *Salt spray test.*—A salt spray test shall be made in accordance with Specification QQ–M–151 to determine corrosion resistance and adequacy of platings.

"4.4.8 *Enclosure test.*—The enclosure test shall be conducted as specified in Standard MIL–STD–108."

Custodians:
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